

Soccermatics Project 2 — Expected Danger Model

Objective

The aim of this project was to construct an **Expected Danger (xD)** model to evaluate how likely a pass is to eventually lead to a goal. This mirrors concepts behind modern expected threat models and helps quantify creative contribution beyond assists.

We implemented a two-stage model using **StatsBomb event data from the 2015/16 Premier League season**.

Methodology

Stage 1: $P(\text{shot}|\text{pass}) \rightarrow$ Logistic Regression

Stage 2: $P(\text{goal}|\text{shot},\text{pass}) \rightarrow$ Linear Regression

Combined: $\text{xD} = \text{Stage 1} \times \text{Stage 2}$

1. Identify passes leading to shots (15-second window)

For every shot, we labeled passes by the same team within the previous 15 seconds as “*danger passes*” (target = 1). All other passes were labeled 0.

2. Probability of a shot after a pass — Logistic Regression

$$P(\text{shot} \mid \text{pass})$$

Features used:

- Start and end x, y pass coordinates
- Distance from end of pass to goal
- Squared distance term (to capture non-linear spatial effects)

These simple transformations allow the model to learn that danger increases sharply as passes end closer to goal.

3. Probability of a goal given a shot — Linear Regression

$$P(\text{goal} \mid \text{shot}, \text{pass})$$

Following the assignment instructions, we used linear regression on the subset of passes that led to shots to estimate the probability that the resulting shot was a goal. (Note: in practice, logistic regression would be used here.)

4. Expected Danger Calculation

$$xD = P(\text{shot} \mid \text{pass}) \times P(\text{goal} \mid \text{shot}, \text{pass})$$

5. Player Aggregation

For each player:

- Sum xD over all passes
- Convert to per-90 rate:

$$xD/90 = \frac{90 \cdot \sum xD}{\text{minutes played}}$$

- Count **danger passes per 90** (passes leading to a shot)

Players were also grouped by position for fair comparison.

xD vs. Danger Passes — what each measures

Danger passes reward any pass that actually results in a shot, which makes them partially outcome-driven and sensitive to teammate decisions and finishing. **xD** instead scores every pass based on the *probability* of creating a goal, making it a better indicator of **creative pass quality and intent**, not just the frequency of shot-leading events.

Results

Top 10 Players (Overall)

	player	team	90s	position	main_position	xD/90	xD/90_Rank	Danger_Passes/90	Danger_Passes/90_Rank	Rank_Difference
367	Mesut Özil	Arsenal	35.566788	Center Attacking Midfield	Attacking Midfield/Winger	0.560712	1	9.475132	1	0
454	Santiago Cazorla González	Arsenal	15.347575	Left Defensive Midfield	Midfielder	0.542579	1	9.903845	1	0
122	David Josué Jiménez Silva	Manchester City	20.938095	Center Attacking Midfield	Attacking Midfield/Winger	0.509601	2	8.119172	2	0
162	Francesco Fabregas i Soler	Chelsea	34.593404	Right Defensive Midfield	Midfielder	0.488952	2	6.417408	2	0
139	Eden Hazard	Chelsea	26.321309	Left Wing	Attacking Midfield/Winger	0.464974	3	6.192701	7	-4
2	Aaron Ramsey	Arsenal	31.059851	Left Defensive Midfield	Midfielder	0.404945	3	6.374789	3	0
296	Kevin De Bruyne	Manchester City	23.692897	Center Attacking Midfield	Attacking Midfield/Winger	0.394417	4	7.597214	4	0
27	Alexis Alejandro Sánchez Sánchez	Arsenal	28.599387	Left Wing	Attacking Midfield/Winger	0.392313	5	6.363773	6	-1
185	Gnégneri Yaya Touré	Manchester City	27.403704	Right Defensive Midfield	Midfielder	0.386456	4	5.875118	5	-1
238	Jesús Navas González	Manchester City	27.100444	Right Wing	Attacking Midfield/Winger	0.383274	6	5.276666	17	-11

The model successfully identifies known elite creators from the 2015/16 season. Mesut Özil's league-leading 19 assists align with his top Expected Danger (xD) per 90 rating (0.56), while players like David Silva and Cesc Fàbregas—renowned for chance creation—rank highly. This suggests the model captures meaningful creative contribution beyond raw assist counts.

Position-Level Analysis

Median values of xD per 90 for each position, shows attacking midfielders closely followed by midfielders and full backs led the xD metric.

xD/90 Danger_Passes/90		
main_position		
Attacking Midfield/Winger	0.237041	4.055084
Midfielder	0.221425	3.052412
Full Back	0.210353	2.622011
Center Forward	0.141389	2.270621
Center Back	0.084414	1.095267
Goalkeeper	0.026531	0.322281

Top Players by Position

Top 10 AM/Wingers by xD

	player	team	90s	position	main_position	xD/90	xD/90_Rank	Danger_Passes/90	Danger_Passes/90_Rank	Rank_Difference
367	Mesut Özil	Arsenal	35.566788	Center Attacking Midfield	Attacking Midfield/Winger	0.560712	1	9.475132	1	0
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27	Alexis Alejandro Sánchez Sánchez	Arsenal	28.599387	Left Wing	Attacking Midfield/Winger	0.392313	5	6.363773	6	-1
238	Jesús Navas González	Manchester City	27.100444	Right Wing	Attacking Midfield/Winger	0.383274	6	5.276666	17	-11
90	Christian Dannemann Eriksen	Tottenham Hotspur	34.483513	Left Wing	Attacking Midfield/Winger	0.379549	7	7.887827	3	4
530	Willian Borges da Silva	Chelsea	32.498859	Right Wing	Attacking Midfield/Winger	0.371425	8	5.692508	12	-4
288	Juan Manuel Mata García	Manchester United	34.647823	Right Wing	Attacking Midfield/Winger	0.365855	9	4.300414	28	-19
132	Dimitri Payet	West Ham United	30.659730	Left Wing	Attacking Midfield/Winger	0.341123	10	7.077688	5	5

Top 10 Midfielders by xD: Cazorla, Fàbregas, Ramsey, Yaya Touré top.

	player	team	90s	position	main_position	xD/90	xD/90_Rank	Danger_Passes/90	Danger_Passes/90_Rank	Rank_Difference
454	Santiago Cazorla González	Arsenal	15.347575	Left Defensive Midfield	Midfielder	0.542579	1	9.903845	1	0
162	Francesc Fàbregas i Soler	Chelsea	34.593404	Right Defensive Midfield	Midfielder	0.488952	2	6.417408	2	0
2	Aaron Ramsey	Arsenal	31.059851	Left Defensive Midfield	Midfielder	0.404945	3	6.374789	3	0
185	Gnégénéri Yaya Touré	Manchester City	27.403704	Right Defensive Midfield	Midfielder	0.386456	4	5.875118	5	-1
54	Bastian Schweinsteiger	Manchester United	14.134098	Left Defensive Midfield	Midfielder	0.380633	5	3.820548	21	-16
264	Jordan Brian Henderson	Liverpool	15.273388	Right Center Midfield	Midfielder	0.338753	6	4.845029	8	-2
369	Michael Carrick	Manchester United	23.322108	Right Center Midfield	Midfielder	0.338610	7	3.130077	32	-25
158	Fernando Luiz Rosa	Manchester City	32.396017	Left Defensive Midfield	Midfielder	0.329365	8	4.383255	11	-3
222	James Philip Milner	Liverpool	28.831080	Right Center Midfield	Midfielder	0.328828	9	6.035154	4	5
193	Harry Arter	AFC Bournemouth	19.305872	Left Center Midfield	Midfielder	0.306821	10	4.299210	12	-2

Top 10 Full Backs by xD

	player	team	90s	position	main_position	xD/90	xD/90_Rank	Danger_Passes/90	Danger_Passes/90_Rank	Rank_Difference
50	Bacary Sagna	Manchester City	29.037129	Right Back	Full Back	0.347996	1	4.167079	2	-1
17	Aleksandar Kolarov	Manchester City	27.195461	Left Back	Full Back	0.334667	2	3.971251	5	-3
174	Gaël Clichy	Manchester City	12.820326	Left Back	Full Back	0.331653	3	3.042044	25	-22
47	Ashley Young	Manchester United	12.626283	Right Back	Full Back	0.310137	4	3.247195	16	-12
541	Àngel Rangel Zaragoza	Swansea City	21.584407	Right Back	Full Back	0.308292	5	3.567390	12	-7
203	Ignacio Monreal Eraso	Arsenal	38.518535	Left Back	Full Back	0.296059	6	3.712498	9	-3
15	Alberto Moreno Pérez	Liverpool	30.465270	Left Back	Full Back	0.291777	7	5.120585	1	6
319	Leighton Baines	Everton	17.608820	Left Back	Full Back	0.285596	8	3.861701	7	1
403	Pablo Javier Zabaleta Girod	Manchester City	12.573218	Right Back	Full Back	0.284112	9	2.863229	27	-18
312	Kyle Naughton	Swansea City	21.992720	Right Back	Full Back	0.280102	10	3.592098	10	0

☐ Sagna & Kolarov rank highly

Top Strikers by xD

	player	team	90s	position	main_position	xD/90	xD/90_Rank	Danger_Passes/90	Danger_Passes/90_Rank	Rank_Difference
522	Wayne Mark Rooney	Manchester United	28.313244	Center Forward	Center Forward	0.253336	1	3.743831	1	0
464	Sergio Leonel Agüero del Castillo	Manchester City	27.588074	Center Forward	Center Forward	0.219086	2	3.081041	8	-6
130	Diego da Silva Costa	Chelsea	28.247246	Center Forward	Center Forward	0.207044	3	2.761331	13	-10
187	Graziano Pellè	Southampton	26.037720	Center Forward	Center Forward	0.196272	4	3.341306	3	1
438	Romelu Lukaku Menama	Everton	37.657061	Center Forward	Center Forward	0.194129	5	2.947654	9	-4
110	Daniel Andre Sturridge	Liverpool	11.591427	Center Forward	Center Forward	0.189928	6	3.192014	6	0
277	José Leonardo Ulloa	Leicester City	12.104434	Right Center Forward	Center Forward	0.187777	7	2.726274	14	-7
508	Troy Deeney	Watford	38.988972	Right Center Forward	Center Forward	0.181590	8	3.334276	4	4
528	Wilfried Guemiand Bony	Manchester City	14.648905	Center Forward	Center Forward	0.176579	9	2.662315	15	-6
89	Christian Benteke Liolo	Liverpool	18.525382	Center Forward	Center Forward	0.175582	10	3.184820	7	3
194	Harry Kane	Tottenham Hotspur	39.525031	Center Forward	Center Forward	0.169866	11	3.415557	2	9
131	Dieumerci Mbokani Bezua	Norwich City	19.231729	Center Forward	Center Forward	0.169031	12	2.599870	16	-4
398	Olivier Giroud	Arsenal	28.844791	Center Forward	Center Forward	0.162807	13	3.224152	5	8

Rooney exceptionally high on both metrics.

Striker rankings largely reflect role. Target-men like Giroud and forwards like Kane often contribute through hold-up play, movement, and drawing defenders — actions that support attacks but don't always raise immediate shot probability — so this model under-credits them. By contrast, strikers like Costa, who more directly assist shot-producing moments, tend to score higher in this framework.

Top 10 CB's by xD

Daily Blind extremely high here and unsurprising when looking at his progressive passing numbers throughout his career.

	player	team	90s	position	main_position	xD/90	xD/90_Rank	Danger_Passes/90	Danger_Passes/90_Rank	Rank_Difference
105	Daley Blind	Manchester United	36.725468	Left Center Back	Center Back	0.209695	1	2.641219	1	0
351	Martín Gastón Demichelis	Manchester City	12.289061	Right Center Back	Center Back	0.164420	2	1.708837	8	-6
331	Mamadou Sakho	Liverpool	22.105479	Left Center Back	Center Back	0.151810	3	2.035694	3	0
393	Nicolás Hernán Otamendi	Manchester City	31.586576	Right Center Back	Center Back	0.151389	4	1.392997	19	-15
516	Vincent Kompany	Manchester City	13.966011	Right Center Back	Center Back	0.145072	5	1.432048	18	-13
315	Laurent Koscielny	Arsenal	33.699720	Left Center Back	Center Back	0.142805	6	1.632061	11	-5
250	John Stones	Everton	33.130400	Right Center Back	Center Back	0.132461	7	1.328085	22	-15
46	Ashley Williams	Swansea City	38.606768	Left Center Back	Center Back	0.132078	8	1.657740	10	-2
166	Gabriel Armando de Abreu	Arsenal	21.418178	Right Center Back	Center Back	0.131931	9	1.260611	25	-16
501	Toby Alderweireld	Tottenham Hotspur	40.442211	Right Center Back	Center Back	0.128650	10	2.299578	2	8

Top 10 GK's by xD

```
filtered_df[filtered_df['main_position'] == 'Goalkeeper'].sort_values('xD/90_Rank').head(10).drop(columns = {'player_id', 'total_minutes', 'xD'})
```

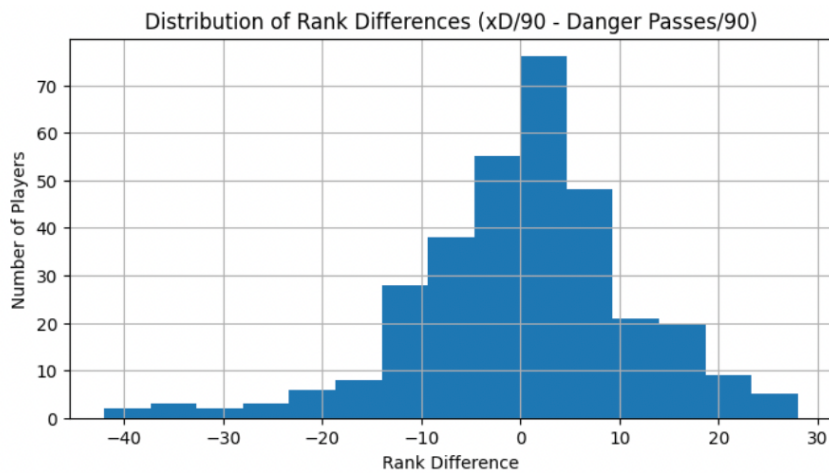
	player	team	90s	position	main_position	xD/90	xD/90_Rank	Danger_Passes/90	Danger_Passes/90_Rank	Rank_Difference
432	Rob Elliot	Newcastle United	22.445527	Goalkeeper	Goalkeeper	0.032244	1	0.400971	9	-8
197	Heurelho da Silva Gomes	Watford	40.176358	Goalkeeper	Goalkeeper	0.030906	2	0.373354	11	-9
164	Fraser Forster	Southampton	19.283631	Goalkeeper	Goalkeeper	0.030736	3	0.466717	5	-2
56	Ben Foster	West Bromwich Albion	16.079376	Goalkeeper	Goalkeeper	0.030466	4	0.124383	26	-22
184	Glyn Oliver Myhill	West Bromwich Albion	24.656563	Goalkeeper	Goalkeeper	0.029490	5	0.283900	19	-14
63	Bradley Guzan	Aston Villa	30.017959	Goalkeeper	Goalkeeper	0.028874	6	0.266507	22	-16
521	Wayne Hennessey	Crystal Palace	31.028805	Goalkeeper	Goalkeeper	0.027891	7	0.322281	14	-7
417	Petr Čech	Arsenal	36.241252	Goalkeeper	Goalkeeper	0.027630	8	0.275929	20	-12
518	Vito Mannone	Sunderland	20.453590	Goalkeeper	Goalkeeper	0.027575	9	0.195565	23	-14
98	Costel Fane Pantilimon	Sunderland	18.300672	Goalkeeper	Goalkeeper	0.027513	10	0.601071	2	8

```
203]: filtered_df['Rank_Difference'].std()
```

```
203]: 11.074753035565244
```

The rank difference between xD/90 and danger passes per 90 has a standard deviation of ~11 ranks, indicating that xD meaningfully differentiates pass quality from volume.

Players like Memphis Depay and Jesse Lingard ranked much higher in danger pass volume than xD, while Jeffrey Schlupp showed the opposite pattern. Nicely illustrates trade-off between quality versus quantity of those passes that help their team.



```
filtered_df.nsmallest(5, 'Rank_Difference')[['player', 'position', 'xD/90_Rank', 'Danger_Passes/90_Rank']]
```

	player	position	xD/90_Rank	Danger_Passes/90_Rank
366	Memphis Depay	Left Wing	23	65
31	Ander Herrera Agüera	Center Attacking Midfield	11	51
379	Morgan Schneiderlin	Right Defensive Midfield	21	56
361	Matteo Darmian	Right Back	19	53
237	Jesse Lingard	Right Wing	25	58

```
filtered_df.nlargest(5, 'Rank_Difference')[['player', 'position', 'xD/90_Rank', 'Danger_Passes/90_Rank']]
```

	player	position	xD/90_Rank	Danger_Passes/90_Rank
234	Jeffrey Schlupp	Left Back	48	20
6	Adam Johnson	Right Wing	40	13
227	Jan Kirchhoff	Center Defensive Midfield	56	29
53	Bamidele Alli	Center Attacking Midfield	45	20
458	Sebastian Prödl	Right Center Back	54	30

Limitations & Future Improvements

The logistic model produced a ROC-AUC of ~ 0.50 , and because none of the predicted probabilities exceeded the standard 0.5 threshold, every pass was classified as “no shot.” However, the **probability values still varied meaningfully across passes**, and using these raw probabilities (rather than the binary predictions) allowed the model to produce **sensible and intuitive xD rankings**.

This model is intentionally simple for instructional purposes. Key limitations:

- No **cross-validation** and models are fit and evaluated on the same data
- **Linear regression** used for a probability task (assignment requirement)
- **Extremely limited feature set** — the model treats all passes equally regardless of context. It cannot differentiate between a corner kick and an open play pass, a counter-attack and slow build-up, or whether defenders are pressing. This makes it

naive compared to real-world models that incorporate game state, defensive pressure, pass type, and tempo.

Potential extensions:

- Logistic model for $P(\text{goal} \mid \text{shot})$
- Tree-based models (XGBoost / Random Forest). These would better capture the spatial logic of football. For example, the center of the goal is at $(x=120, y=40)$, so danger peaks when y is close to 40, not when y is simply high or low. This non-linear relationship between y -coordinate and scoring probability is naturally captured by decision trees splitting on optimal coordinate ranges, whereas linear models struggle with this geography.
- Cross-validation and out-of-sample evaluation
- **Add passing context & freeze-frame information** — pass height, set pieces vs open play, defensive pressure, counter-attack indicators, number of defenders between ball and goal

Alternative Models

Unconditional Model: An unconditional model could estimate the expected goals value (xG) generated by each pass, rather than just binary goal outcomes. For passes that lead to shots, the target would be the xG of that shot; for passes that don't lead to shots, the target would be zero. This would provide more stable estimates than binary goals while still capturing the full attacking contribution of a pass. However, this still treats all non-shot passes as equal (zero value), regardless of whether they advanced possession dangerously. Additionally, using pre-existing xG models (rather than training your own) would create feature overlap issues, as xG models already incorporate many of the same pass-context features (counter-attacks, defensive pressure, etc.), leading to double-counting of these factors.

xT: A more comprehensive approach would model the change in possession value or threat (similar to Expected Threat/xT frameworks), which would reward passes that advance into dangerous areas even without immediately generating shots. This would better capture deep-lying playmakers who progressively build attacks over multiple actions. However, such models require defining zone-based value functions or modelling longer action sequences, adding significant complexity.

Conclusion

The Expected Danger model provides a simple but intuitive measure of creative threat in football. Despite its simplicity, it successfully identifies elite creators and highlights the value of spatial pass features in predicting attacking outcomes. Future improvements could bring the model closer to modern football analytics pipelines, but this exercise builds a strong foundation for understanding event-based expected value models.